Grade: Check Plus

Assignment 1

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a. $P(X \geq 60)$ ✓ Qa

0.45

b. Expected Value and Standard Deviation

$E(X) = \Sigma\, x \cdot f(x)$ ✓ Qb

Total 1.00 50.00 3190.00

$E(X) = \$50$

$Var(X) = E(X^2) - [E(X)]^2 = 3190 - (50)^2 = 3190 - 2500 = 690$

$\sigma = \sqrt{690} \approx \$26.27$

c. Response to the Manager ✓ Qc

The manager is confusing the *expected value* (a long-run average) with an *observable outcome*. The expected value of $50 does not mean any individual customer will withdraw exactly $50 — in fact, no one can. The expected value is a weighted average of all possible outcomes. If we observed thousands of withdrawals, the average amount withdrawn would converge to $50.

d. P(within one standard deviation of the mean)

One standard deviation range: $\mu \pm \sigma = 50 \pm 26.27 \rightarrow (23.73, 76.27)$ ✓ Qd

The values of X that fall within this interval are: 40, 60

$P(23.73 < X < 76.27) = f(40) + f(60) = 0.30 + 0.15 = 0.45$

e. P(within two standard deviations of the mean)

Two standard deviation range: $\mu \pm 2\sigma = 50 \pm 52.54 \rightarrow (-2.54, 102.54)$

All values in the table (10, 20, 40, 60, 80, 100) fall within this interval.

$P = 0.10 + 0.15 + 0.30 + 0.15 + 0.25 + 0.05 = 1.00$ ✓ Qe